

DATA 266 Lab 1 — Team 32 Report

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Repository: <https://github.com/sakotiya/data266-lab1>

Team ownership statement

Each of us designed, implemented, trained, and evaluated our own models in our own folders. We kept our code and experiment results separate, then compared the results after both members had finished their runs.

Before training, we agreed on the main shared decisions. We interpreted 100K/10K as the number of sequences, used non-overlapping data slices for Task 1, built vocabularies from training data only, and used the same `metrics_report.csv` format. This made it easier to place our results side by side. The comparison sections below summarize what we found together.

Individual contributions

Shreya's contribution

I built and trained the 12-layer deep-narrow character GPT for Task 1. I also trained three Yelp sentiment models for Task 2: a mean-pooling baseline, a TextCNN, and a BiLSTM. For Task 3, I trained a CycleGAN with UNet generators and PatchGAN discriminators. My work is organized around configuration-driven notebooks, and I prepared the preprocessing, training logs, metrics, generated samples, and failure analyses for my models.

Zoheb's contribution

I built and trained the 4-layer shallow-wide character GPT for Task 1. For Task 2, I trained a BiLSTM mean-pooling baseline, a TextCNN, and an attention-based BiLSTM. For Task 3, I trained a CycleGAN with ResNet-9 generators and 70 × 70 PatchGAN discriminators. I organized my work using Python source modules and notebooks, and prepared the checkpoints, metrics, plots, logs, and error reviews for my models.

We agreed to use the same main evaluation format so that the results could be compared fairly.

Task 1 — GPT-style character language model from scratch

1.1 Shared setup

Both models follow the same rules, so differences between them come from the design choices in §1.2.

- **Data.** TinyStories, character-level. Both members read the same raw file at **different, non-overlapping offsets**: Shreya chars [0, 34M), Zoheb chars from 34M. Each member therefore has their own train/validation split.

- **Split size.** "Training (100K) and validation (10K)" is read as **sequences** of 256 characters: 100,000 train sequences (25.6M characters) and 10,000 validation sequences (2.56M characters). Targets are inputs shifted by one.
- **Vocabulary.** Own `char_to_idx / idx_to_char`, built from the training split only, with an `<unk>` slot for unseen characters.
- **No prebuilt Transformer modules.** Attention (Q/K/V projections, scaled dot product, causal mask, softmax, head split/merge), LayerNorm placement, feed-forward, residuals, token and positional embeddings and the LM head are all hand-written. Neither code base uses `nn.Transformer*`, `nn.MultiheadAttention` or fused scaled-dot-product attention.
- **Causal mask verified behaviourally.** Both members changed the token at position `j` and confirmed that the logits at every position before `j` are bit-identical, while logits from `j` onward change.
- **Training.** Cross-entropy loss, AdamW ($\beta = 0.9$, 0.95, weight decay 0.1), linear warm-up followed by cosine decay, gradient clipping at 1.0, 10 epochs.
- **Generation.** Greedy decoding and temperature sampling from the prompt "Once upon a time".

1.2 Model comparison — architecture and hyperparameters

The two models are the team's depth-vs-width comparison: a deep-narrow model against a shallow-wide one at the same embedding width and the same number of training tokens.

	shreya_akotiya	zoheb_waghu
Design idea	deep-narrow	shallow-wide
Transformer blocks	12	4
Attention heads (dims/head)	8 (32)	4 (64)
Embedding width <code>d_model</code>	256	256
Feed-forward width	1024 (4x)	1024 (4x), GELU
Normalisation	pre-LN	pre-LN
Positional encoding	learned	learned
Context length	256	256
Vocabulary size	78	97
Dropout	0.05	0.1
Parameters	9,570,816	3,274,752
Batch size (tokens/step)	64 (16,384)	32 (8,192)
Steps (per epoch / total)	1,562 / 15,620	3,125 / 31,250
Peak → min learning rate	2.5e-4 → 2.5e-5	3e-4 → 3e-5
Warm-up	781 steps (5%)	1,000 steps (~3%)
Epochs	10	10

	shreya_akotiya	zoheb_waghu
Training tokens seen	256M	256M
Seed	1337	1337
Hardware	NVIDIA Tesla T4 16 GB (Google Colab), CUDA, PyTorch 2.11	NVIDIA GeForce RTX 4090 24 GB, AMD Ryzen 9 7950X, CUDA, PyTorch 2.5.1

The two models also differ in dropout, batch size, learning rate and head size, not only in depth. The comparison is therefore **deep-narrow vs shallow-wide as complete designs**, not a controlled ablation of depth alone.

1.3 Metrics — side by side

All numbers are from each member's `metrics_report.csv` (team schema) and `metrics_report_extended.csv`. Cross-entropy is in nats per character on each member's own 10K-sequence validation split.

Metric	shreya_akotiya	zoheb_waghu
Training cross-entropy (final epoch)	0.6038	0.6982
Validation cross-entropy (best)	0.6327	0.7019
Perplexity	1.883	2.018
Bits-per-character	0.913	1.013
Generalization gap (val – train)	0.0289	0.0037
Top-1 next-character accuracy	79.9%	77.7%
Distinct-1 / 2 / 3 (character) ‡	0.0029 / 0.0286 / 0.1247	0.0125 / 0.0997 / 0.2832
Repeated 4-gram rate (character) ‡	0.721	0.559
Gradient norm mean / max	0.687 / 17.79	0.744 / 9.77
Loss spikes †	22	0
NaN / Inf events	0	0
Parameter count	9,570,816	3,274,752
Training throughput (tokens/sec) ¶	24,209 (T4)	663,173 (RTX 4090)
Generation throughput (tokens/sec) ¶	87.6	507.6
Peak GPU memory (<code>torch.cuda.max_memory_allocated</code>)	8.06 GB	0.99 GB
Total training time ¶	176.2 min	6.4 min
Best epoch	10 (last)	10 (last)

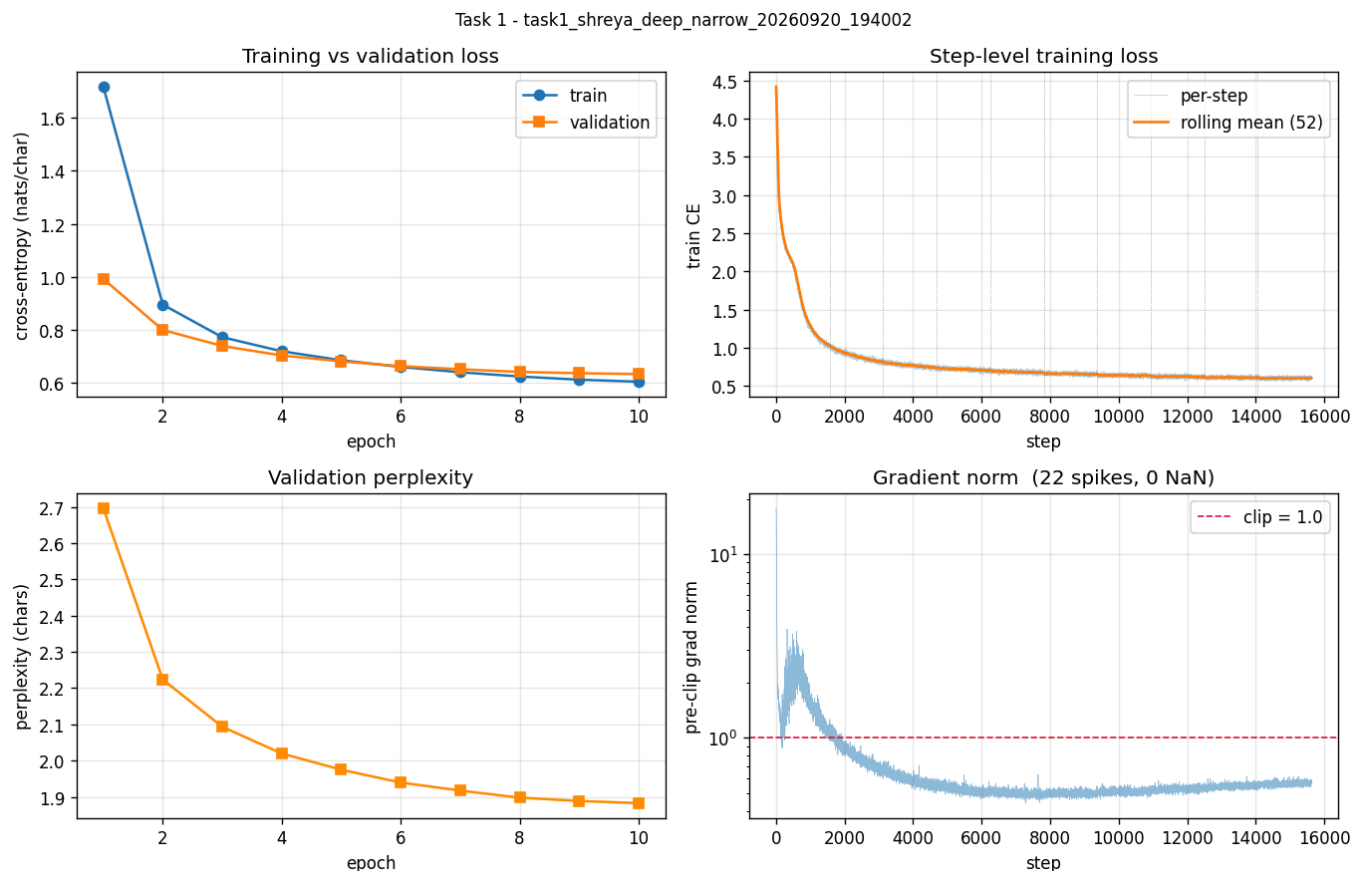
Comparability notes

- ‡ **Diversity metrics are not directly comparable.** Both use the same formula (unique n-grams ÷ total n-grams over characters), but Shreya's are measured on 20,000 generated characters at $T = 1.0$ and Zoheb's on ~3,600 characters pooled over greedy, $T = 0.8$ and $T = 1.2$. Distinct-n falls and repeated-n-gram rate rises as the text gets longer, so the gap between the columns mostly reflects sample length, not model quality.

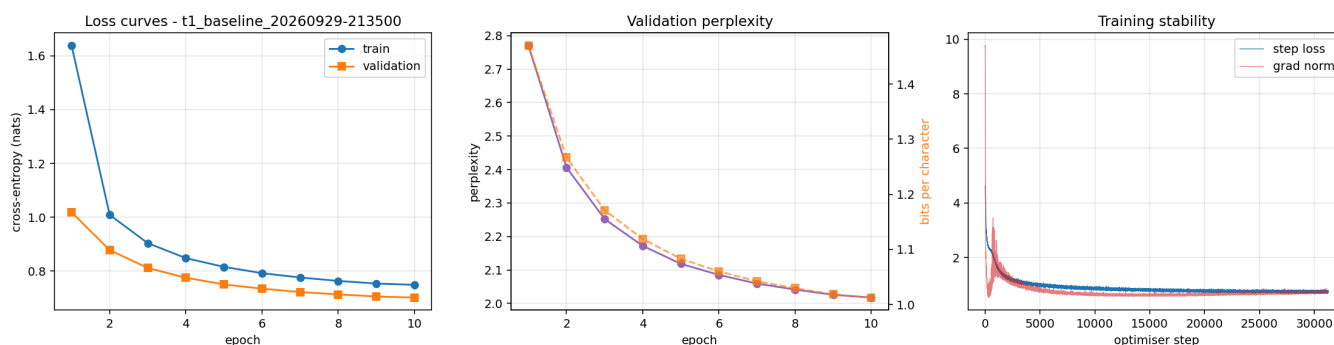
- † **Loss spikes use different detectors.** Shreya flags a step whose loss exceeds the rolling mean by 3σ (100-step window); her 22 spikes are each about 7% above the local mean. Zoheb flags a loss above $2\times$ the running median, a much looser threshold that none of Shreya's spikes would meet. Both runs are stable by either definition: no NaNs, and validation loss improved every epoch.
- ¶ **Speed is not comparable.** Shreya trained on a Tesla T4 and Zoheb on an RTX 4090, a much faster GPU, so throughput and training time mostly reflect the hardware, not the models.
- **Peak memory is comparable.** Both runs measure it with `torch.cuda.max_memory_allocated`. The $8\times$ difference comes from the deeper model (12 vs 4 blocks of stored activations) and the larger batch (64 vs 32).
- The two validation sets are different slices of TinyStories, so the loss comparison is close but not paired.

1.4 Training curves

shreya_akotiya (12 × 256):



zoheb_waghu (4 × 256):



Validation loss per epoch:

Epoch	1	2	3	4	5	6	7	8	9	10
shreya — val bpc	1.431	1.153	1.066	1.014	0.982	0.956	0.939	0.924	0.917	0.913
zoheb — val bpc	1.470	1.267	1.171	1.119	1.083	1.060	1.042	1.030	1.019	1.013

Both curves fall at every epoch and both models reach their best validation loss at the last epoch. The deeper model is ahead from epoch 1, and the gap between the two models grows from 0.04 bpc at epoch 1 to about 0.10 bpc by epoch 10.

1.5 Joint analysis

What we noticed from the comparison

1. **The deeper model performed better on next-character prediction.** The 12-layer model reached 0.913 bits per character and 79.9% next-character accuracy, compared with 1.013 bits per character and 77.7% accuracy for the 4-layer model. The difference makes sense for this task because a character model has to build characters into words, words into phrases, and phrases into sentences. The extra layers give it more steps to do that.
2. **The deeper model also costs more.** It has about 2.9 times as many parameters and used about eight times more peak GPU memory. The two training times cannot be compared directly, because Shreya's run used a Tesla T4 while Zoheb's used an RTX 4090.
3. **The smaller model had a smaller generalization gap, but that does not mean it learned more.** Its gap was 0.0037 compared with 0.0289 for the deeper model. The smaller gap mostly reflects the lower capacity of the shallow model: it fit the training data less closely as well as the validation data.
4. **Neither model had fully converged after 10 epochs.** Validation loss was still improving at the end of both runs, so the final epoch was still the best checkpoint. Ten epochs was the training limit, not the point where either model had completely finished learning.
5. **Both runs were stable.** Neither model produced NaN or Inf values. The largest gradient norm, 17.8, occurred at the beginning of the deeper model's training. After warm-up, its maximum was 2.9, and no step after step 1,910 needed clipping.

Strengths

- Both implementations pass a behavioural causal-mask test, not only a visual one.

- Both produce fluent, correctly spelled English at low temperature, with correct word boundaries, punctuation, quotation marks and the typical TinyStories structure.
- Disjoint data slices give each member a genuinely separate train/validation split.

Weaknesses and shared failure modes

The two models fail in the same ways, which suggests the failures come from the shared setup (character-level next-token prediction, 256-character context, small model, no decoding constraints) rather than from either design.

Failure type	shreya_akotiya example	zoheb_waghu example
Repetition under greedy decoding	repeats "Once upon a time, there was a little girl named Lily."	"The box was so happy and the box was so happy to see the box."
Semantic contradiction / impossible meaning	"Lily was happy to hear that her mom was sad."	"Her pocket seemed sad to melt."
Losing track of who is who	"Lily didn't want to share his toy friends"	restarts a new story mid-generation ("Once upon a time...")
Non-words at high temperature	"strets", "pumpins", "knowled" (3.0% non-words at T = 1.2)	"aboven", "MPleaf", "spreadying" at T = 1.2

In both models, temperature moves the problem from one type of failure to another. Greedy and low-temperature decoding usually give cleaner spelling, but the model can repeat common phrases. Higher temperatures reduce repetition, but they also increase spelling and meaning errors. In Shreya's temperature sweep, T = 0.8 gave the best balance: the non-word rate was only 0.42% and no repeated phrase was detected. Zoheb saw the same general pattern: his repeated 4-gram rate decreased as the temperature increased, but the semantic problems became more noticeable at T = 1.2.

Limitations

- Single seed per model, so there is no variance estimate.
- The two models differ in several hyperparameters at once (see §1.2), so the quality gap cannot be attributed to depth alone.
- The diversity and loss-spike metrics were computed with different sample sizes and detectors (see §1.3).
- The 256-character context is shorter than most stories, so neither model can keep a story consistent from start to finish.
- Generation uses no KV cache, so throughput figures understate what the models could do.

What the team would try next

1. **Train longer.** Both curves were still falling. We expect 15–20 epochs to lower validation loss further for both, with the deep model's gap growing first.
2. **Controlled depth ablation.** Train 256×4 and 256×12 with the same batch size, learning rate, dropout and seed, to isolate the effect of depth.
3. **Top-k / nucleus sampling ($p \approx 0.9$).** Cutting the low-probability tail should keep the diversity of higher temperatures while avoiding their non-words, and should break greedy loops.

4. **Longer context (512–1024)** and masking at story boundaries, targeting the coherence and story-restart failures that decoding changes cannot fix.
5. **Unified evaluation script.** Re-score both models' diversity on the same number of characters at the same temperature, with one loss-spike definition.

1.6 Individual failure analyses

Full write-ups with verbatim snippets are in each member's folder:

- `task1_llm/shreya_akotiya/failure_analysis.md`:
 1. semantic contradiction (greedy);
 2. pronoun and reference confusion ($T = 0.5$);
 3. non-words at high temperature ($T = 1.0$);
 4. the temperature trade-off across greedy to $T = 1.2$.
- `task1_llm/zoheb_waghu/failure_analysis.md`:
 1. phrase-level repetition loop (greedy);
 2. invalid words and semantic breakdown at high temperature ($T = 1.2$);
 3. story-boundary confusion: restarting a new story mid-generation (greedy).

1.7 Evidence

	shreya_akotiya	zoheb_waghu
Run ID	task1_shreya_deep_narrow_20260920_194002	t1_baseline_20260929-213500
Config	task1_llm/shreya_akotiya/config.yaml	task1_llm/zoheb_waghu/configs/gpt_baseline.yaml
Checkpoint	task1_llm/shreya_akotiya/checkpoints/task1_shreya_deep_narrow_20260920_194002_best.pt (sha256 f540786b...)	task1_llm/zoheb_waghu/checkpoints/t1_baseline_20260929-213500_best.pt (sha256 6b7ecfe1...)
Raw log	reproducibility/raw_logs/shreya_akotiya/task1_llm/task1_shreya_deep_narrow_20260920_194002.log	reproducibility/raw_logs/zoheb_waghu/task1_llm/t1_baseline_20260929-213500.log (+ .jsonl)
Manifest	reproducibility/manifests/shreya_akotiya/task1_shreya_deep_narrow_20260920_194002.json	reproducibility/manifests/zoheb_waghu/task1_llm_manifest.md
Loss curves	outputs/plots/loss_curves.png	outputs/plots/loss_curves_t1_baseline_20260929-213500.png
Samples	outputs/samples/{greedy,T0.5,T0.8,T1.0,T1.2}.txt	outputs/samples/samples_t1_baseline_20260929-213500.txt
Causal-mask check	outputs/plots/causal_mask_check.png	src/evaluate.py::causal_mask_probe (logged in run)
Metrics	metrics_report.csv, metrics_report_extended.csv	metrics_report.csv, metrics_report_extended.csv

Task 2 — Yelp Polarity sentiment classification

2.1 Shared setup

- **Dataset: Yelp Polarity** (confirmed), with 560,000 training and 38,000 test reviews, binary and exactly balanced.
- **Same test set.** Both members evaluate on the **full official 38,000-review test split**, touched once, at the end. Because the test set is identical, the test metrics below are directly comparable across members, unlike Task 1, where each member had their own validation slice.
- **Validation is held out from the training split** in both pipelines. Each member's three models share one split and one vocabulary, so the paired McNemar tests within a member are valid.
- **No pretrained embeddings or language models.** Every model uses an `nn.Embedding` initialised randomly and trained end to end.
- **Negation words are kept.** Both members remove stopwords but exempt negation words (`not`, `no`, `never`, `n't` forms), because removing them would turn "not good" into "good".
- **Same metric set.** Both pipelines report accuracy, precision/recall/F1 (macro, micro and weighted), confusion matrix, ROC-AUC, PR-AUC, MCC, Brier score, ECE, 95% bootstrap CIs, paired McNemar against the member's baseline, per-slice macro-F1 and error rate, parameter count, training time, examples/sec and peak GPU memory.

2.2 Preprocessing — side by side

	shreya_akotiya	zoheb_waghu
Training data used	all 540,000 (+ 20,000 validation)	89,997 subsample (+ 9,999 validation)
Test	full 38,000	full 38,000
Malformed rows dropped	0 train / 0 test (none found; 35 reviews empty after cleaning, kept as padding)	4 train / 0 test
Lowercase, strip punctuation/special characters	yes	yes
Contractions	apostrophe removed (<code>don't</code> → <code>dont</code>), kept as a negation word	expanded before stopword removal (<code>don't</code> → <code>do not</code>)
Stopword list	sklearn (318); 13 negation words exempted → 305 removed	NLTK (198); 39 negation words exempted → 158 removed
Word normalisation	light suffix stemming (<code>-s</code> , <code>-ed</code> , <code>-ing</code> , <code>-ly</code> , ...)	WordNet lemmatisation
Vocabulary	48,436 words (min frequency 5, cap 50K), 99.47% token coverage	30,000 words (min frequency 2, cap 30K), 1.28% test OOV
Max length (tokens)	200: keeps 96.6% of reviews whole	256: keeps 97.8% of reviews whole
Embedding dimension	200 (all three models)	128 (M1, M2) / 256 (M3)

EDA findings both members recorded: the data is perfectly balanced, and review length is right-skewed (a long tail of reviews over 500 words). Shreya also found that **negative reviews are longer** (median 112 vs 84 words).

2.3 Models — architecture and hyperparameters

Each member trained one baseline and two experimental models. No two of the six share an architecture and hyperparameter set.

	Model	Encoder	Pooling	Params	Dropout	Optimiser / LR	Batch / epochs
Shreya	Baseline — mean-pool	none (bag of embeddings)	masked mean	9,687,602	0.3	Adam 1e-3	256 / 3
	Exp. 1 — TextCNN	Conv1d, widths 3/4/5 × 100 filters	max over time	9,928,102	0.5	Adam 1e-3	256 / 3
	Exp. 2 — BiLSTM	1-layer BiLSTM, hidden 128	masked max over time	10,025,634	0.3	Adam 1e-3	256 / 3
Zoheb	Baseline — BiLSTM-mean	1-layer BiLSTM, hidden 128	masked mean	4,104,449	0.3	Adam 1e-3	64 / 8 (early stop)
	Exp. 1 — TextCNN	Conv1d, widths 2/3/4/5 × 128 filters	max over time	4,135,681	0.5	Adam 1e-3, wd 1e-5	64 / 8 (early stop)
	Exp. 2 — BiLSTM-attention	2-layer BiLSTM, hidden 256	additive (Bahdanau) attention	10,441,217	0.4	AdamW 2e-3, cosine, 300-step warm-up	32 / 12 (early stop)

Design logic.

- **Shreya's set adds sequence information step by step at a fixed embedding size:** no word order (mean-pool), then local n-grams (TextCNN), then full-sequence state (BiLSTM).
- **Zoheb's set starts from a sequential baseline** and tests two changes: replacing recurrence with convolutions (TextCNN), and replacing mean pooling with learned attention (BiLSTM-attn).
- **Model selection.** Both members keep the checkpoint with the best validation score. Every model peaked within the first one to three epochs: Shreya's at epochs 2–3 of 3, and Zoheb's early stopping chose epoch 1–2.

2.4 Metrics — all six models on the same 38K test set

Metric	S: mean-pool	S: TextCNN	S: BiLSTM	Z: BiLSTM-mean	Z: TextCNN	Z: BiLSTM-attn
Accuracy	0.9314	0.9435	0.9485	0.9346	0.9357	0.9402
Precision (macro)	0.9315	0.9435	0.9485	0.9346	0.9359	0.9404
Recall (macro)	0.9314	0.9435	0.9485	0.9346	0.9357	0.9402
F1 (macro)	0.9314	0.9435	0.9485	0.9346	0.9357	0.9402
F1 (micro)	0.9314	0.9435	0.9485	0.9346	0.9357	0.9402
F1 (weighted)	0.9314	0.9435	0.9485	0.9346	0.9357	0.9402
ROC-AUC	0.9793	0.9863	0.9891	0.9830	0.9836	0.9853
PR-AUC	0.9787	0.9866	0.9894	0.9835	0.9840	0.9856

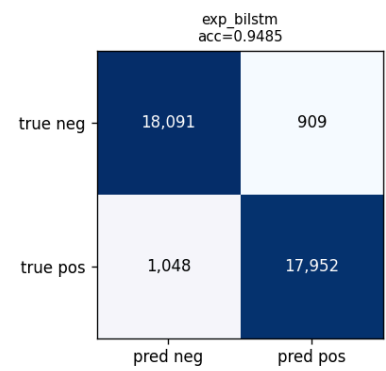
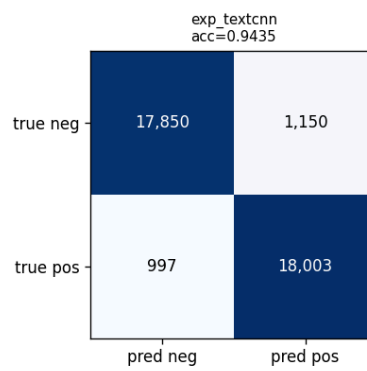
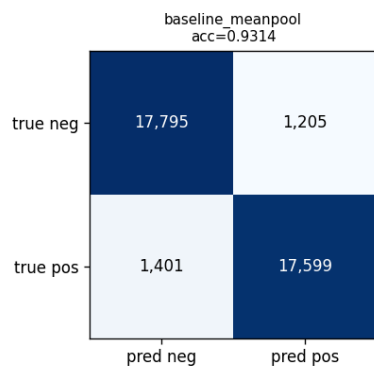
Metric	S: mean-pool	S: TextCNN	S: BiLSTM	Z: BiLSTM-mean	Z: TextCNN	Z: BiLSTM-attn
MCC	0.8629	0.8870	0.8970	0.8693	0.8716	0.8806
Brier score	0.0516	0.0430	0.0384	0.0489	0.0481	0.0449
ECE ‡	0.0046	0.0111	0.0086	0.0134	0.0054	0.0088
Confusion TN / FP	17,795 / 1,205	17,850 / 1,150	18,091 / 909	17,766 / 1,234	17,963 / 1,037	18,046 / 954
Confusion FN / TP	1,401 / 17,599	997 / 18,003	1,048 / 17,952	1,250 / 17,750	1,405 / 17,595	1,317 / 17,683
Accuracy 95% CI	[0.9289, 0.9339]	[0.9412, 0.9458]	[0.9463, 0.9507]	[0.9321, 0.9371]	[0.9333, 0.9383]	[0.9379, 0.9426]
Macro-F1 95% CI	[0.9289, 0.9339]	[0.9412, 0.9458]	[0.9463, 0.9507]	[0.9321, 0.9371]	[0.9333, 0.9383]	[0.9379, 0.9426]
MCC 95% CI	[0.8578, 0.8679]	[0.8824, 0.8915]	[0.8926, 0.9014]	[0.8642, 0.8741]	[0.8667, 0.8767]	[0.8760, 0.8855]
McNemar p vs own baseline	–	1.6e-24	9.4e-55	–	0.369 (n.s.)	2.6e-7
Parameters	9.69M	9.93M	10.03M	4.10M	4.14M	10.44M
Training time ¶	41 s	248 s	350 s	19 s	17 s	436 s
Examples/sec ¶	39,502	6,547	4,637	18,569	21,723	825
Peak GPU memory	338 MB	460 MB	2,286 MB	1,280 MB	219 MB	1,526 MB
Hardware	Tesla T4	Tesla T4	Tesla T4	RTX 4090	RTX 4090	RTX 4090

Micro-F1 equals accuracy for single-label classification. Macro and weighted F1 are almost identical to it because the test set is exactly balanced.

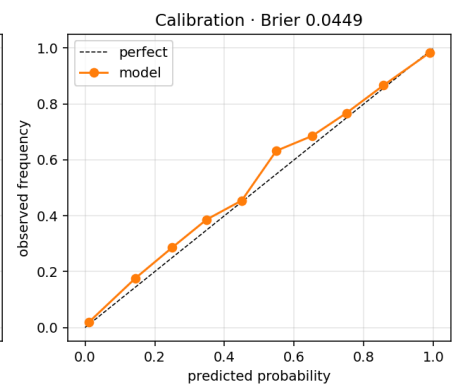
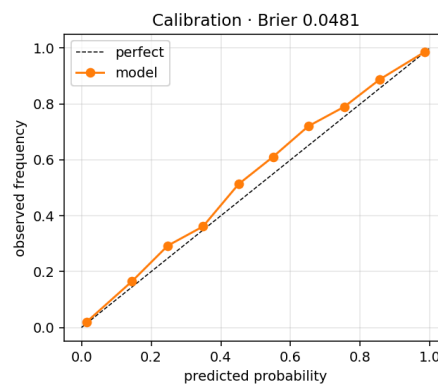
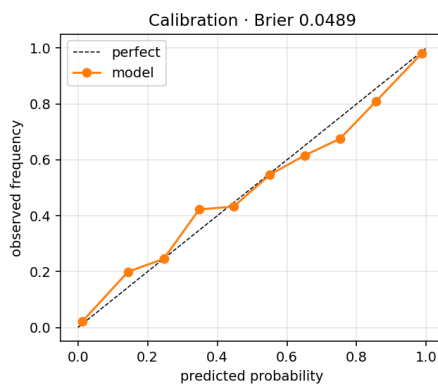
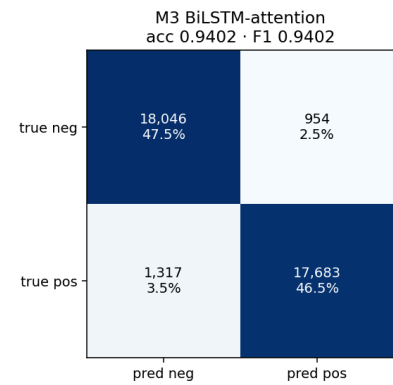
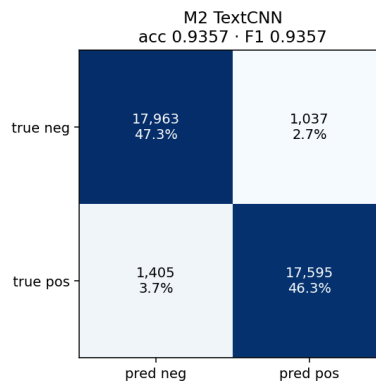
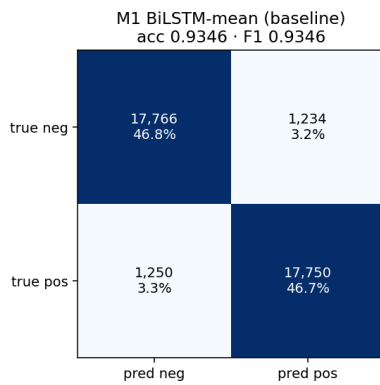
Comparability notes

- ¶ **Speed is not comparable across members.** Shreya trained on a Tesla T4 (Google Colab) and Zoheb on an RTX 4090. Training time also depends on training-set size (540K vs 90K reviews) and epoch count. Peak memory is comparable: both use `torch.cuda.max_memory_allocated`.
- ‡ **ECE uses different bin counts** (10 bins for Shreya, 15 for Zoheb), so small ECE differences across members are not meaningful. Brier score has no such parameter and is directly comparable.
- Bootstrap CIs use 1,000 resamples (Shreya) and 2,000 (Zoheb). This affects the CI estimates only slightly.

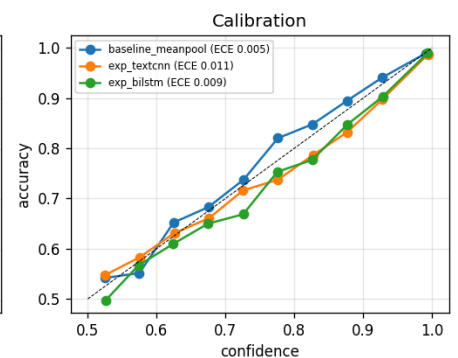
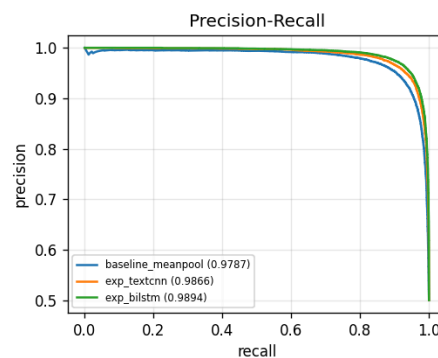
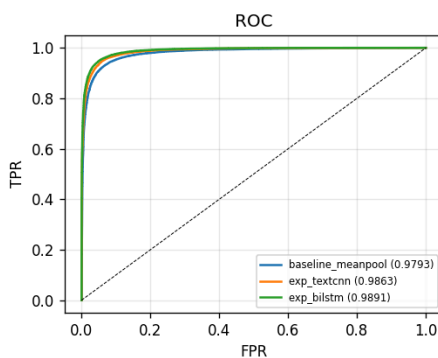
Confusion matrices (Shreya):



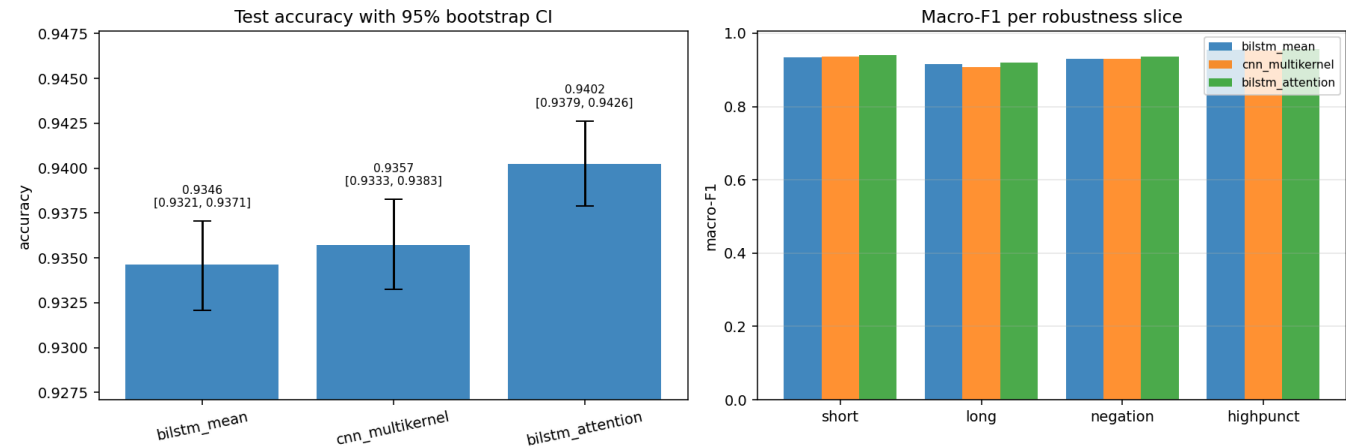
Confusion matrices and calibration (Zoheb):



ROC, PR and calibration curves (Shreya):



Model comparison (Zoheb):



2.5 Per-slice robustness

Each member defined their own slices, so the numbers within a member's set are comparable, and across members only the trends are.

Shreya (lengths in processed tokens; negation detected after preprocessing). Each cell is macro-F1 / error rate.

Slice (n)	mean-pool	TextCNN	BiLSTM
≤ 40 tokens (16,731)	0.931 / 6.8%	0.943 / 5.6%	0.949 / 5.0%
41–100 tokens (14,542)	0.932 / 6.8%	0.945 / 5.4%	0.949 / 5.1%
> 100 tokens (6,727)	0.924 / 7.2%	0.934 / 6.3%	0.941 / 5.6%
has negation (28,544)	0.924 / 7.4%	0.940 / 5.9%	0.945 / 5.3%
no negation (9,456)	0.927 / 5.3%	0.930 / 5.0%	0.936 / 4.6%

Zoheb (lengths in tokens; negation detected in the raw review). Each cell is macro-F1 / error rate.

Slice (n)	BiLSTM-mean	TextCNN	BiLSTM-attn
short, ≤ 50 tokens (18,860)	0.935 / 6.4%	0.936 / 6.3%	0.941 / 5.9%
long, > 200 tokens (1,697)	0.915 / 7.5%	0.908 / 8.0%	0.919 / 7.1%
contains negation (22,583)	0.929 / 6.6%	0.931 / 6.4%	0.936 / 5.9%
≥ 3 exclamation marks (6,944)	0.955 / 4.4%	0.952 / 4.7%	0.958 / 4.1%

Shared findings.

- **Long reviews are the weakest slice for all six models.** Long reviews are more often mixed (praise and complaint in one review), and a few lose their tail to truncation. Each member's strongest model (Shreya's BiLSTM, Zoheb's BiLSTM-attn) stays the most accurate on long reviews, although it is not

the one whose score drops least. In both sets the TextCNN drops most (Shreya: +0.7–0.9 error points; Zoheb: 0.936 → 0.908 macro-F1), consistent with fixed-width filters losing long-range contrast.

- **Reviews with a negation are harder for every model**, but keeping negation words limits the damage. The penalty is largest for the model with no word order: Shreya's mean-pool error rate rises 2.1 points from the no-negation to the negation slice, against 0.7–0.8 points for her TextCNN and BiLSTM. Zoheb's models lose about 0.5 macro-F1 points on the negation slice relative to their overall score.
- **Strongly emotional reviews (≥ 3 exclamation marks) are the easiest slice** in Zoheb's analysis (error rate 4.1–4.7%), because they carry unambiguous sentiment.

2.6 Joint analysis

What we learned from the comparison

1. **The BiLSTM trained by Shreya had the highest test accuracy at 94.85%.** Its confidence interval did not overlap Zoheb's best BiLSTM-attention interval. However, architecture alone cannot explain the difference. Shreya's models used 540K training reviews, while Zoheb used 90K, and the vocabulary and embedding sizes were also different. The difference in training-set size is probably the most important factor.
2. **Keeping word order helped.** Shreya's mean-pooling baseline was the weakest of her three models at 93.14%, while both the TextCNN and BiLSTM performed better. Zoheb's baseline was already a BiLSTM, which is why it started at 93.46% even with less training data.
3. **TextCNN was especially useful compared with a bag-of-words model.** Shreya's TextCNN improved over her mean-pooling baseline by 1.2 percentage points, and the McNemar test confirmed that the improvement was significant. In Zoheb's experiments, however, TextCNN was not significantly better than his BiLSTM baseline. This suggests that local phrase features help most when the baseline does not understand word order at all.
4. **Attention pooling improved Zoheb's BiLSTM.** The attention BiLSTM improved over the mean-pooling BiLSTM by 0.56 percentage points. It also required much more computation. Shreya's BiLSTM masked max-pooling model reached a higher score, but the two results are not a perfectly controlled pooling comparison because the training data and embedding sizes were different.
5. **The probability outputs were generally useful.** All models had reasonably low Brier scores and ECE values. Within each member's experiments, the model with the best accuracy also had the best Brier score.
6. **Most models reached their best validation result early.** The models usually peaked within the first few epochs. This suggests that future improvements should focus more on the amount of data, input length, and model design than simply adding many more epochs.

Strengths

- All six models exceed 93% accuracy with embeddings learned entirely from scratch.
- Every claim is backed by a significance test (CIs plus paired McNemar), not by point estimates alone.
- The negation handling is measured: both members report a negation slice, and the bag-of-words model's larger negation penalty shows why negations had to be kept.

Weaknesses and limitations

- **Confounded comparison across members.** Training-set size, vocabulary, embedding size, preprocessing (stemming vs lemmatisation) and hardware all differ between the two sets, so cross-member differences cannot be attributed to any single factor.
- **Single seed per model**, so there is no estimate of run-to-run variance.
- **Slices were defined independently** (different length cut-offs and negation detectors) and ECE bins differ, so those numbers are not fully comparable across members.
- **Label noise.** Both error reviews found reviews whose text clearly contradicts their gold label (rating-text mismatch), which caps achievable accuracy.
- **Truncation** at 200 / 256 tokens cuts the end of the ~2–3% longest reviews, where the final verdict often sits.

What the team would try next

1. **A controlled cross-member test.** Train Zoheb's BiLSTM-attn on Shreya's full 540K split (or Shreya's BiLSTM on Zoheb's 90K split) to separate the effect of data size from the effect of architecture.
2. **A paired McNemar test between the two best models.** Both members scored the same 38K test reviews, so a paired test of Shreya's BiLSTM against Zoheb's BiLSTM-attn is possible once the prediction files are aligned by row.
3. **A hierarchical or sentence-level encoder** for long, mixed reviews. Both error reviews identify mixed sentiment as a leading error type, and the long-review slice is the weakest for all six models.
4. **Shared evaluation code.** Use the same slice definitions, ECE bins and bootstrap count, so every robustness number is comparable.
5. **Multiple seeds** (e.g. 3 per model) to put variance bars on the close calls, such as Zoheb's TextCNN vs his baseline.

2.7 Individual error reviews

Each member manually reviewed 20 errors from one of their own models: 5 confident false positives, 5 confident false negatives, 5 near-threshold errors and 5 slice-specific errors. Each error has a type and a testable fix.

	shreya_akotiya	zoheb_waghu
Model reviewed	BiLSTM (best, 94.85%)	BiLSTM-mean baseline (93.46%)
Slice used for the last 5	reviews containing a negation	long reviews (> 200 tokens), the worst slice
Most common error types	mixed sentiment / split verdicts, temporal shifts ("EDIT: now much better"), negation scope (negation of a competitor, a requirement or a past state), sarcasm	mixed sentiment (12 of 20) , rating-text mismatch (2), domain term (2), length truncation (2), negation (1), sarcasm (1)
Proposed fix	phrase- or aspect-level sentiment aggregation; parse negation scope; recency weighting	chunked hierarchical sentence encoder over up to 512 tokens, measured on long-review macro-F1
Full write-up	task2_sentiment/shreya_akotiya/failure_analysis.md	task2_sentiment/zoheb_waghu/failure_analysis.md, outputs/error_review_t2_m1_baseline.md

Both reviews reach the same conclusion from different models. The remaining errors are mostly **discourse-level**: mixed or changing sentiment within one review, sarcasm, and negations whose scope is not the review's subject. A single pooled vector cannot represent these. Both members also found label noise, which no model can fix.

2.8 Evidence

	shreya_akotiya	zoheb_waghu
Run IDs	task2_shreya_main_20260921_222011 (all three models)	t2_m1_baseline_20260929-211318, t2_m2_cnn_20260929-211446, t2_m3_bilstm_attn_20260929-211521
Config	task2_sentiment/shreya_akotiya/config.yaml	task2_sentiment/zoheb_waghu/configs/(_shared.yaml + one per model)
Checkpoints	task2_sentiment/shreya_akotiya/checkpoints/{baseline_meanpool,exp_textcnn,exp_bilstm}.pt	task2_sentiment/zoheb_waghu/checkpoints/t2_m{1,2,3}_..._best.pt
Raw logs	reproducibility/raw_logs/shreya_akotiya/task2_sentiment/task2_shreya_main_20260921_222011.log	reproducibility/raw_logs/zoheb_waghu/task2_sentiment/t2_m*_20260929-*.log (+ .jsonl)
Manifest	reproducibility/manifests/shreya_akotiya/task2_shreya_main_20260921_222011.json	reproducibility/manifests/zoheb_waghu/task2_sentiment_manifest.md
Metrics	metrics_report.csv, metrics_report_extended.csv, outputs/slice_metrics.csv, outputs/mcnemar.csv	metrics_report.csv, metrics_report_extended.csv
Plots	outputs/plots/{eda,confusion_matrices,curves}.png	outputs/plots/{eda_overview,model_comparison}.png, outputs/confusion_matrices/confusion_and_calibration.png
Error review	failure_analysis.md, outputs/error_review.md	failure_analysis.md, outputs/error_review_t2_m1_baseline.md

Task 3 — CycleGAN image style transfer (Monet ↔ photo)

3.1 Shared setup

- **Data.** Kaggle gan-getting-started: 300 Monet paintings and 7,038 photographs, unpaired, 256 × 256. Both members trained at full 256px resolution, with the same augmentation: resize to 286, random 256 crop, horizontal flip.
- **Direction names.** This report uses the instructor's convention: **A = Monet, B = photo**. So **A2B = Monet → photo** and **B2A = photo → Monet** (the Kaggle direction). Shreya's notebook internally uses A = photo, so her outputs/pred_A2B/ folder holds photo → Monet. All tables below use the instructor's names.
- **Same CycleGAN recipe.** Both models have two generators and two discriminators, trained with an LSGAN adversarial loss, cycle-consistency loss ($\lambda = 10$ in both directions), identity loss ($0.5 \times \lambda = 5$), a 50-image fake pool for each discriminator, Adam ($\text{lr } 2\text{e-}4$, $\beta = (0.5, 0.999)$), batch size 1 with instance

normalisation, and one epoch = one pass over the 7,038 photos (each Monet image is reused about 23 times per epoch).

- **Same evaluation.** All image-quality numbers follow the instructor's `Part3_Evaluation_Script.ipynb`: the first 300 sorted images per set, Inception-v3 features, FID and MiFID per direction, and submission value = mean of the two directions. Both members then computed KID, precision/recall, density/coverage, cycle L1, LPIPS and content cosine on the same 300 images, so these numbers are directly comparable.
- **Leaderboard integrity.** Every submitted value comes from each member's own CycleGAN inference. Pretrained networks (Inception-v3, AlexNet for LPIPS) are used only to *measure* images, never to generate or edit them.

3.2 Model comparison — architecture and hyperparameters

The two models differ mainly in the **generator**: Shreya used a UNet with skip connections from pix2pix, and Zoheb used the ResNet-9 generator from the CycleGAN paper.

	shreya_akotiya	zoheb_waghu
Generator	UNet : 7 stride-2 encoder blocks down to a 2×2 bottleneck, 6 decoder blocks + output layer, skip connection at every level, dropout 0.5 in 3 decoder blocks	ResNet-9 : 7×7 conv, 2 downsamples, 9 residual blocks at 64×64 , 2 upsamples, reflection padding
Generator parameters	41,821,187 each	11,378,179 each
Discriminator	PatchGAN: 3 stride-2 convs ($64 \rightarrow 256$) + output conv; 31×31 score map	70×70 PatchGAN: 3 stride-2 convs + a stride-1 512 conv; score map
Discriminator parameters	662,593 each	2,764,737 each
Total parameters	84,967,560	28,285,832
Epochs (steps)	80 (563,040)	40 (281,520)
LR schedule	constant to epoch 61, linear decay over epochs 62–80 (ends at $1.9e-5$)	constant 20 epochs, linear decay to 0 over 20
Seed	42	1337
Hardware	NVIDIA A100-SXM4-40GB (Google Colab), PyTorch 2.11	NVIDIA RTX 4090 24 GB, PyTorch 2.5.1
Training time	9.66 h	6.69 h

Design logic.

- **Shreya (UNet)**: skip connections pass edges and layout straight from the encoder to the decoder, so the generator only has to change colour and texture rather than rebuild the scene. The expected strength is content preservation; the expected risk is that the skip paths make it easy to copy the input or hide information in it.
- **Zoheb (ResNet-9)**: nine residual blocks repeatedly transform texture and colour at 64×64 . This is the generator from the original CycleGAN paper, with a larger 70×70 patch discriminator focused on local brush texture.

- Like Task 1, this compares **two complete designs**. Generator, discriminator, epoch count, schedule, seed and GPU all differ, so a difference cannot be credited to the generator alone.

3.3 Metrics — side by side (same 300 images per direction)

Metric	Shreya A2B (Monet → photo)	Shreya B2A (photo → Monet)	Zoheb A2B (Monet → photo)	Zoheb B2A (photo → Monet)
FID ↓	102.784	97.904	103.197	98.855
MiFID	0.4206	0.4043	0.4181	0.4047
KID ↓	0.0185	0.0069	0.0182	0.0076
Precision (realism)	0.730	0.540	0.703	0.507
Recall (diversity)	0.447	0.730	0.480	0.680
Density	1.005	0.489	1.013	0.433
Coverage	0.850	0.763	0.897	0.690
Cycle-reconstruction L1 ↓ ([0, 1] pixels)	0.0223	0.0216	0.0332	0.0398
LPIPS, input vs translation (how much changed)	0.244	0.304	0.354	0.378
Content cosine, input vs translation ↑	0.861	0.821	0.797	0.772
Human audit (style / content / artifacts)	pending	pending	pending	pending
Inter-rater agreement (Cohen's κ)	pending	pending	pending	pending

Submission and training	shreya_akotiya	zoheb_waghu
Submission FID / MiFID (mean of both directions)	100.344 / 0.4124	101.026 / 0.4114
Leaderboard score (FID + MiFID) / 2 ↓	50.38	50.72
Kaggle public / private score and rank	pending	pending
Generator loss (final 10% of steps)	2.439	2.970
Discriminator loss, D_A + D_B (final 10%)	0.171	0.251
Cycle loss, unweighted (final 10%)	0.080	0.132
Identity loss, unweighted (final 10%)	0.046	0.097
Generator gradient norm (mean) †	38.40	23.36
NaN / Inf steps	0	0
Images / sec ¶	16.2	11.7
Peak GPU memory †	1.78 GB	12.78 GB

Comparability notes

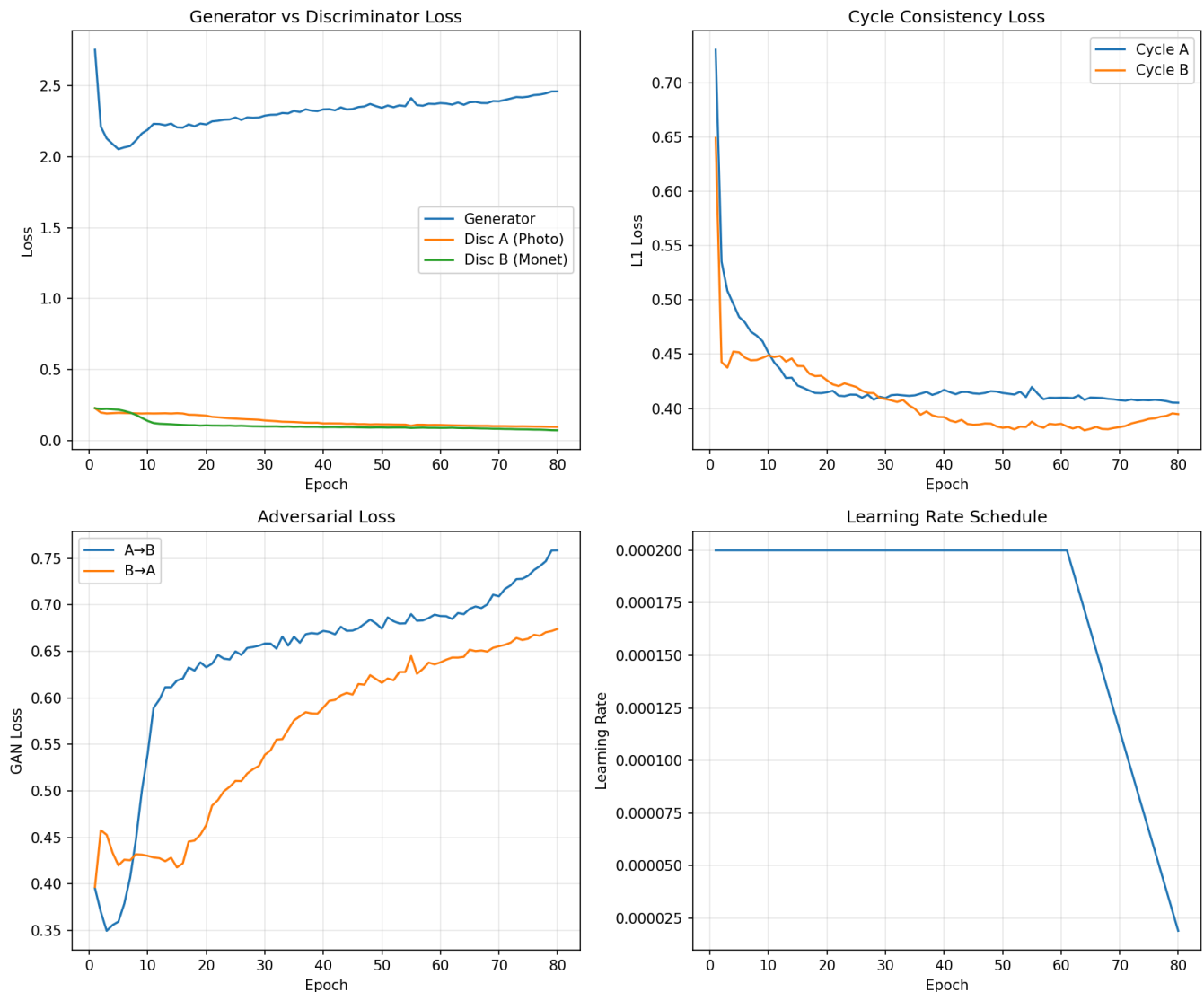
- FID across directions is not comparable.** A2B is compared with real photos and B2A with real Monet paintings, so the two directions have different reference sets. Compare each direction across the two

members, not A2B against B2A.

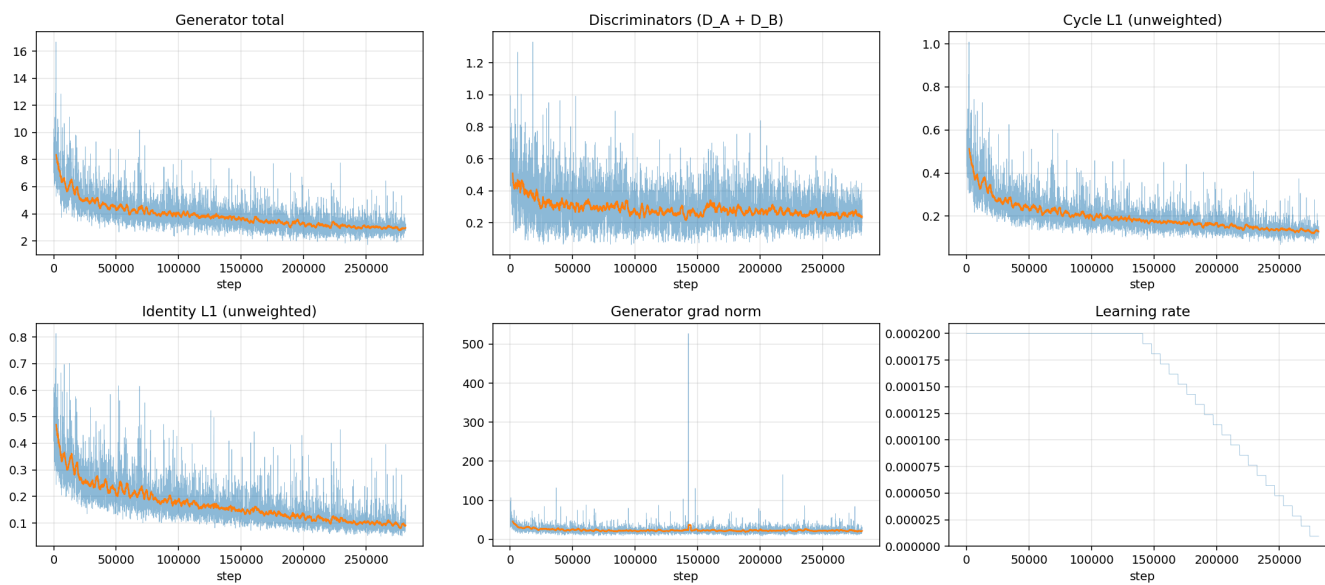
- † **Gradient norm and peak memory were measured differently.** Zoheb's values come from his training run. Shreya's run did not log them, so hers come from **one extra diagnostic epoch** (7,038 steps) started from the epoch-80 checkpoint, with the same losses and data pipeline (run1_grad_diagnostic.ipynb). Her gradient norm therefore describes the trained model, not the average over training.
- ¶ **Speed is not comparable** (A100 vs RTX 4090), and the two models also have very different parameter counts.
- **Training losses** come from different architectures, so they are context, not a ranking.

3.4 Training curves

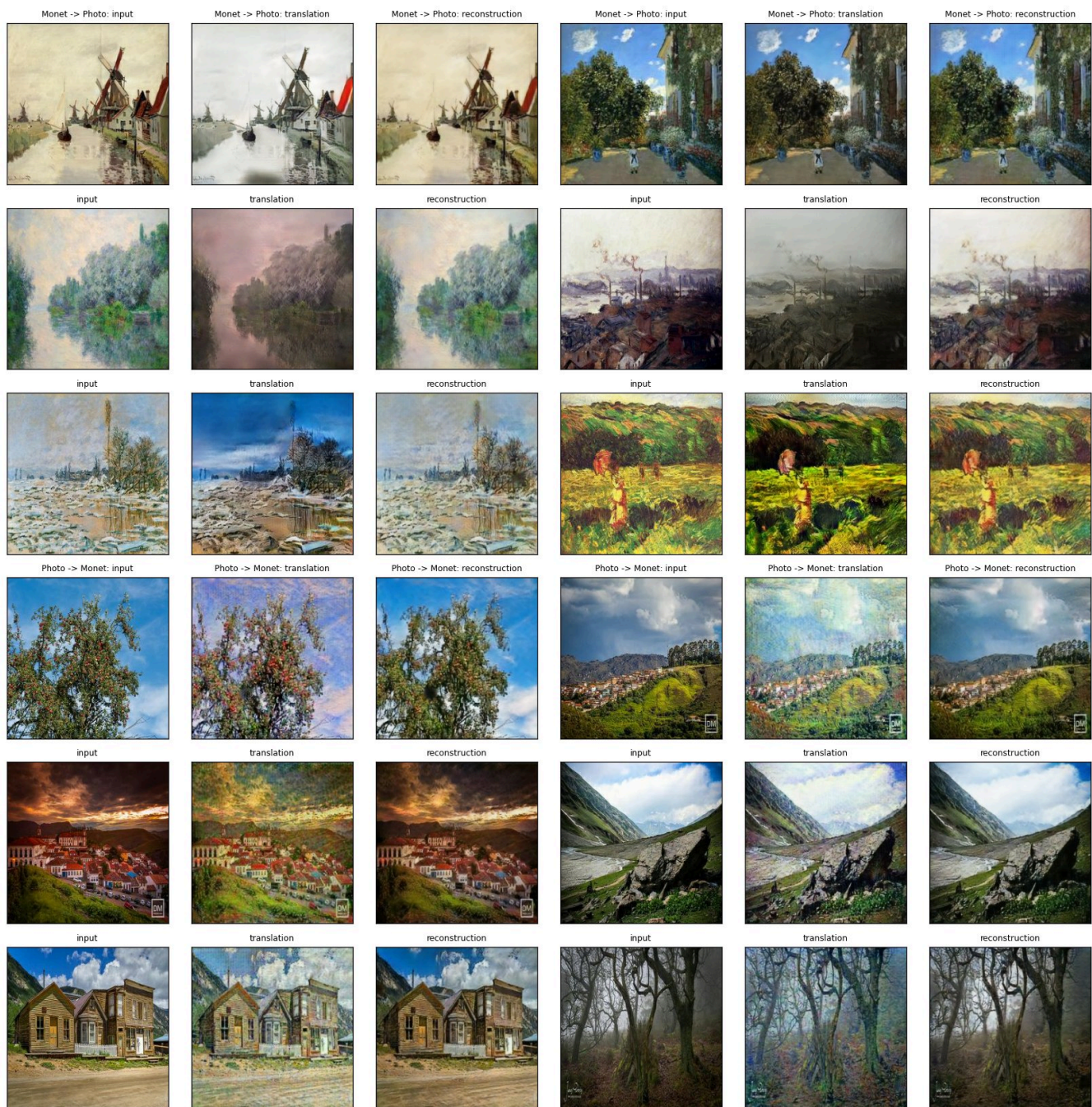
shreya_akotiya (UNet, 80 epochs):



zoheb_waghu (ResNet-9, 40 epochs):



Example translations (zoheb_waghu): input, translation and cycle reconstruction.



Shreya's failure-case images are in `task3_gan/shreya_akotiya/outputs/plots/failure_candidates_{B2A,A2B}.jpg` (source, translation, cycle reconstruction for 15 cases per direction).

What the training shows.

- **Both runs were numerically stable:** 0 NaN or Inf steps. Zoheb's largest gradient spike (527 at epoch 21) recovered by the next logged step without loss divergence.
- **Losses fell fastest early in both runs.** Shreya's cycle loss was flat (about 0.41 weighted) from epoch 30. Zoheb's checkpoint score improved mostly up to epoch 20 and then varied by only 0.46 points to epoch 40.

- **The discriminators gradually got ahead in Shreya's run.** After epoch 5, the photo → Monet adversarial loss rose (0.40 → 0.76) while discriminator loss fell (0.23 → 0.08). In Zoheb's run, the discriminator loss stayed at 0.245–0.310 from epoch 11, which means the discriminators separated real from fake usefully without completely winning.

3.5 Joint analysis

What the comparison shows

1. **Image quality is effectively tied.** The two models are within about one FID point in each direction (B2A 97.9 vs 98.9; A2B 102.8 vs 103.2), and the leaderboard scores are 50.38 vs 50.72. That gap is well inside the noise of 300-image FID: Zoheb's FID varied over a range of 4.7–6.5 points across his checkpoints after epoch 20, and Shreya's v4 run moved 5–8 points between checks 10 epochs apart. Neither model is better on FID or KID.
2. **The UNet preserves content more strongly.** Shreya's model has about half the cycle error (0.022 vs 0.033–0.040), smaller LPIPS change (0.24–0.30 vs 0.35–0.38) and higher content cosine (0.82–0.86 vs 0.77–0.80) in both directions. This is what skip connections are expected to do: they carry the scene structure straight to the output.
3. **But low cycle error is not proof of good translation.** Both members found translations that look badly broken but still reconstruct almost perfectly. Shreya's night photos become a tiled blue-grey pattern yet have the *lowest* cycle errors (0.013–0.021). Zoheb's purple-cloud photo becomes a bright paint field, yet the reconstruction restores it. The source information is carried in subtle signals a person cannot see ("steganography", Chu et al., 2017). UNet skip connections make this easier, so part of the UNet's cycle-error advantage comes from hidden information, not from faithful translation.
4. **Both models fail in the same direction-specific way. Photo → Monet is diverse but less convincing:** recall is higher than precision for both (Shreya 0.73 / 0.54; Zoheb 0.68 / 0.51), so the outputs cover much of the Monet style space but many don't look like a real Monet. **Monet → photo is convincing but narrow:** precision is higher than recall for both (0.73 / 0.45; 0.70 / 0.48), so the outputs look photographic but cover only part of the variety of real photos. The shared cause is the data: only 300 Monet paintings, against 7,038 photos.
5. **More capacity and training did not buy a better score.** Shreya's model has 3× the parameters and trained twice as many epochs, for a 0.34-point lower leaderboard score, which is within noise. Zoheb's checkpoint sweep shows the score had nearly plateaued by epoch 20, and Shreya's three alternative runs (v2–v4: DiffAugment, lower identity weight, removed outer skip, EMA) all scored worse (54.2–55.9). The limiting factors appear to be the small Monet set and the noisy evaluation, not model size.

Strengths

- Both members built a complete CycleGAN, trained it at full 256px resolution, and produced a valid submission from their own model's inference.
- All quality metrics use the instructor's method on the same 300 images, so the side-by-side comparison is fair.
- Cycle consistency is verified with measured reconstruction distances, not only assumed from the loss term, and both members inspected the cases where low cycle error hides bad output.

Weaknesses and shared failure modes

Failure type	shreya_akotiya example	zoheb_waghu example
Dark or low-detail scenes break down	night sky → blotchy blue-grey field with a tiled pattern (09fc404e31)	dark sunset → structureless pastel field (02ded12bbd)
Monet → photo becomes too dark / too contrasty	golden Parliament sunset → almost black (b1ea5d5a7d)	the same image → black and orange extremes (b1ea5d5a7d)
Too little change (still looks like the source domain)	office building stays photographic (063ab57d41)	the same building, mild purple tint only (063ab57d41)
The same coastal painting fails (a619072f82)	rainbow-like horizontal band across the sky, kept in the reconstruction	becomes saturated and high-contrast; the reconstruction is washed out
Colour shifts	sunset loses its red horizon (02ded12bbd)	orange clouds turn cyan, storm sky turns teal (0845e8dc24, 0254fc91bd)

Several of the same images fail for both models (b1ea5d5a7d, a619072f82, 063ab57d41, 02ded12bbd, plus 6782e7cb2a and 04f59976b5). That suggests these images are hard for CycleGAN in general, not for one architecture.

Specific to the UNet: the tiled pattern on night photos and horizontal sky streaks. **Specific to the ResNet-9:** strong hue shifts in clouds and skies.

Limitations

- **Single seed per model.** Run-to-run variance is not measured, and the checkpoint-to-checkpoint swings above suggest it is several FID points.
- **300-image FID is noisy and biased upward.** It matches the leaderboard, but small differences between models are not meaningful.
- **Not a controlled comparison.** Generator, discriminator, epochs, schedule, seed and GPU all differ.
- **Domain imbalance.** With 300 Monet paintings each seen about 23 times per epoch, the Monet discriminator can memorise them.
- **Pending:** the blinded human audit (30 samples, 2 raters, Cohen's κ) and the Kaggle public and private score and rank, for both members.

What the team would try next

1. **Select checkpoints on a held-out validation set** (FID, KID, precision/recall) rather than taking the last epoch, since quality plateaus and fluctuates late in training.
2. **Run three seeds per setting** before treating any FID difference of a few points as real.
3. **Change one thing at a time.** For the UNet: lower only the identity weight (0.5) while keeping the outer skip. For the ResNet-9: a discriminator with a larger receptive field, to target low B2A precision.
4. **Rebalance the discriminators** (e.g. fewer discriminator updates) where they overpower the generator.
5. **Complete the human audit** to check whether people agree with the metric tie, particularly on artifacts.

3.6 Individual failure analyses

Full write-ups with sample IDs, LPIPS and cycle values are in each member's folder:

- task3_gan/shreya_akotiya/failure_analysis.md: 14 failure cases (8 photo → Monet, 6 Monet → photo), the cycle-consistency check including the steganography finding, training stability, shared hard images, and the human-audit protocol. results.md also covers the three alternative runs (v2–v4).

- task3_gan/zoheb_waghu/failure_analysis.md: 18 failure cases (9 per direction: least changed, most changed, worst cycle), cycle-consistency verification, training stability including the epoch-21 gradient spike, and the human-audit protocol.

3.7 Evidence

	shreya_akotiya	zoheb_waghu
Run ID	t3_shreya_unet_128_20261002_023102 (256px; "128" is from an earlier plan)	t3_baseline_20260929-235720
Config	task3_gan/shreya_akotiya/config.yaml	task3_gan/zoheb_waghu/configs/cyclegan_baseline.yaml
Checkpoint	checkpoints/t3_shreya_unet_128_20261002_023102_G_AB_fp16.pt, ..._G_BA_fp16.pt	checkpoints/t3_baseline_20260929-235720_generators_fp16.pt
Raw logs	reproducibility/raw_logs/shreya_akotiya/task3_gan/t3_shreya_unet_128_20261002_023102.log (+_graddiag.{csv,json}, v2-v4 logs)	reproducibility/raw_logs/zoheb_waghu/task3_gan/t3_baseline_20260929-235720.{log,jsonl}
Manifest	reproducibility/manifests/shreya_akotiya/task3_gan_manifest.md, t3_shreya_unet_128_20261002_023102.json	reproducibility/manifests/zoheb_waghu/task3_gan_manifest.md
Metrics	metrics_report.csv, submission.csv	metrics_report.csv, full_metrics_report.csv, submission.csv, outputs/snapshot_metrics.csv
Evaluated images	outputs/pred_A2B/ (photo → Monet, 300), outputs/pred_B2A/ (Monet → photo, 300)	outputs/pred_B2A/ (photo → Monet, 300), outputs/pred_A2B/ (Monet → photo, 300)
Plots	outputs/plots/loss_curves.png, failure_candidates_{B2A,A2B}.jpg	outputs/plots/training_curves_...png, translation_examples.jpg, failure_candidates.jpg
Evaluation code	src/part3_evaluation_shreya.ipynb, src/run1_team_metrics.ipynb, src/human_audit.py	evaluate_local.py, src/snapshot_sweep.py

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